

Separating Mixtures — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, name the **property that differs** between the substances — soluble or insoluble, boiling point, magnetic, particle size — because that property is why the method works.
- The mark for each question is shown in square brackets on the right.
- R_f = distance moved by the substance $\div$ distance moved by the solvent. Both from the baseline. No units.

SECTION 1 – SIMPLE DISTILLATION

1. Sea water is separated by simple distillation to collect pure water.

(a) Describe what happens to the water in the flask when it is heated. [1]

(b) State the purpose of the condenser, and explain how it works. [2]

(c) Give the name of the liquid collected in the receiving flask. [1]

(d) State what is left in the heating flask at the end, and explain why it did not come over with the water. [2]

(e) State the reading on the thermometer while the water is being collected. [1]

2. Explain why the cooling water enters the condenser at the end nearest the receiving flask and leaves at the end nearest the heating flask. [2]

SECTION 2 – CHROMATOGRAPHY

3. Explain why a spot of ink separates into several colours when a chromatogram is run. [3]

4. A chromatogram is run with three known food dyes and one sweet colouring. The solvent front travels **8.0 cm**. E102 travels 2.0 cm, E110 travels 5.6 cm and E133 travels 6.8 cm. The sweet gives spots at 2.0 cm and 6.8 cm.

- (a) Calculate the R_f values of E102 and E133. Show your working. [2]

- (b) State which dyes the sweet contains, and how you know. [2]

- (c) Explain why the sweet colouring cannot be a pure substance. [1]

5. On another chromatogram the solvent front is 12.0 cm from the baseline and a green spot is 9.0 cm from the baseline. Calculate the R_f value of the green dye. [2]

- 6.** A jar contains a dry mixture of salt and sand. Describe, step by step, how to obtain [6]
a sample of dry salt from it. Give a reason for each step.

- 7.** Explain why the steps in the previous question must be done in that order, and [3]
what would go wrong if the filtering and the evaporating were swapped.

- 8.** Name a method that could separate oil from water, and state the property it relies [2]
on.

End of paper. Check every answer has a unit or a form the question actually asked for.